

```
explain code- public class PrintEvenNumbers {
    public static void main(String[] args) {
        int limit = 100;
        System.out.println("Even numbers from 1 to " + limit
+ ":");

        for (int i = 1; i <= limit; i++) {
            if (i % 2 == 0) { // Check if the number is even
                System.out.print(i + " ");
            }
        }
    }
}
```



Let's break down your Java code step by step 📌

◆ Code:

java

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        for (int i = 1; i <= limit; i++) {
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            }
        }
    }
}
```

◆ Explanation:

1. `public class PrintEvenNumbers`
 - Defines a **class** named `PrintEvenNumbers`.
 - In Java, all code must be inside a class.

2. `public static void main(String[] args)`

- This is the **main method**, which is the entry point of every Java program.
 - When you run the program, the execution starts from here.
-

3. `int limit = 100;`

- Declares an **integer variable** `limit` and assigns it the value `100`.
 - This means we want to print even numbers **from 1 to 100**.
-

4. `System.out.println("Even numbers from 1 to " + limit + ":");`

- Prints a message before displaying the numbers.
- Output:

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Even numbers **from 1 to 100**:

5. `for (int i = 1; i <= limit; i++)`

- A **for loop** that runs from `i = 1` to `i = 100`.
 - It increments `i` by 1 each time (`i++`).
-

6. `if (i % 2 == 0)`

- Checks if the current number `i` is **even**.
 - `%` is the **modulus operator** (remainder).
 - If dividing `i` by 2 leaves no remainder (i.e., remainder = 0), then `i` is even.
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7. `System.out.print(i + " ");`

- If the number is even, it prints the number followed by a space.
 - (Uses `print` instead of `println` so numbers appear on the **same line**.)
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◆ Output:

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Even numbers from 1 to 100:

2 4 6 8 10 12 14 16 18 20 ... 100

✓ **Summary:**

This program:

1. Loops through numbers from 1 to 100.
2. Uses `% 2 == 0` to find even numbers.
3. Prints all even numbers in one line.

Would you like me to show a **shorter version** (like skipping odd numbers using `i += 2`) or keep it as-is for beginners?



Is this conversation helpful so far?

